Script-Controlled ACL: Restrict Record Access Based on Field Value

Project Design Phase-II: Technology Stack (Architecture & Stack)

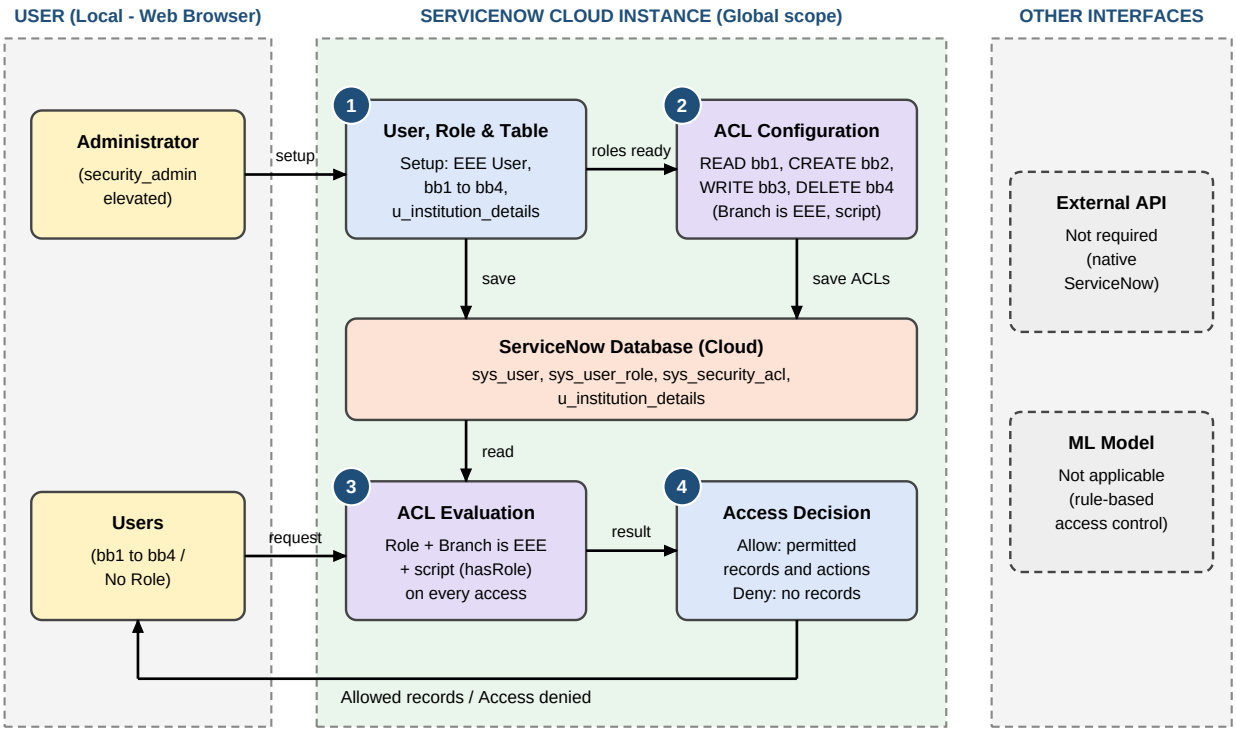
Date: October 3, 2026	**Maximum Marks:** 4 Marks
Team ID: SWTID-2026-8089	**Project Name:** Script-Controlled ACL

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2.

Architecture: Script-Controlled ACL for Record Access Based on Field Value using ServiceNow Access Control (ACL)

Reference: https://www.servicenow.com/docs



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Administrator and users work in the ServiceNow web UI: User Administration, System Definition, System Security → Access Control (ACL), and the Institution Details list (u_institution_details.list).	ServiceNow Web UI (HTML, CSS, JavaScript rendered by the platform), Application Navigator, List and Form views
2.	Application Logic-1	User, role and table setup: the EEE User is created, the roles bb1, bb2, bb3 and bb4 are created and assigned, and the Institution Details table (u_institution_details) is created with its fields and sample records.	ServiceNow User Administration (Users, Roles), System Definition (Tables)
3.	Application Logic-2	ACL configuration: after elevating to security_admin, the READ ACL (bb1, data condition Branch is EEE, advanced script), CREATE ACL (bb2), WRITE ACL (bb3) and DELETE ACL (bb4) are created on u_institution_details.	ServiceNow Access Control (ACL), security_admin role
4.	Application Logic-3	Access evaluation: every time a user opens the list or a record, ServiceNow checks the matching ACLs. The script uses gs.hasRole() to allow admin and bb1 users and returns false for all others, together with the Branch is EEE data condition.	ServiceNow server-side scripting (JavaScript, gs.hasRole), ACL data conditions
5.	Database	Stores the users, roles, role assignments, ACL records and the Institution Details records (Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description).	ServiceNow platform database (tables: sys_user, sys_user_role, sys_user_has_role, sys_security_acl, u_institution_details)
6.	Cloud Database	Database service provided with the ServiceNow cloud instance; no separate database is set up for this project.	ServiceNow cloud-hosted instance database
7.	File Storage	No separate file storage is required; any attachments added to a record are stored with the record.	ServiceNow platform attachments
8.	External API-1	Not required – the complete solution is configured inside ServiceNow without a third-party API.	None (native ServiceNow)
9.	External API-2	Not required.	None
10.	Machine Learning Model	Not applicable – access control is rule-based (role, data condition and script), without a machine learning model.	Not applicable
11.	Infrastructure (Server / Cloud)	Application Deployment on Cloud: ServiceNow cloud instance (Global application scope), accessed through a web browser. Local Server Configuration: Not required – users only need a computer with an internet connection and a web browser. Cloud Server Configuration: ServiceNow-hosted instance with User Administration and Access Control (ACL) available.	ServiceNow Cloud (Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is developed for this project; the solution is configured using the built-in tools of the ServiceNow platform.	ServiceNow Access Control (ACL), scripted ACL (JavaScript), User Administration
2.	Security Implementations	Access to the records is controlled through ServiceNow login, role-based ACLs, the data condition Branch is EEE and the ACL script. Only a user with the elevated security_admin role can create ACLs, users without the role see no records, and the admin keeps full access.	ServiceNow roles and ACLs, security_admin elevation, HTTPS
3.	Scalable Architecture	The same ACLs apply to any number of records and users because access depends on roles and field values. The platform separates the user interface, application logic and database, and the approach can be extended to other branches and tables.	ServiceNow multi-tier platform, roles and ACLs
4.	Availability	The application is hosted on the ServiceNow cloud platform and is available through a web browser to administrators and users without local installation.	ServiceNow cloud-hosted instance
5.	Performance	ACLs are evaluated on the server at the time records are requested, using a simple role check and one data condition, so users see the filtered list without manual steps.	ServiceNow ACL engine (server-side evaluation)

References:

https://c4model.com/
https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
https://www.ibm.com/cloud/architecture
https://aws.amazon.com/architecture
https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d